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## 0.1 Bosonic algebra

$\boxed{1.}$ From the properties of the states $|n\rangle$ we immediately get

$$\hat{N}_a|n\rangle = \hat{a}^\dagger \hat{a}|n\rangle = \hat{a}^\dagger \sqrt{n}|n-1\rangle = n|n\rangle,$$

therefore we see that the state $|n\rangle$ is an eigenstate of $\hat{N}_a$ with eigenvalue $n$. Using the properties of the scalar product we get

$$\begin{aligned}\langle n \,|\, \hat{N}_a \,|\, n\rangle &= \langle n \,|\, \hat{a}^\dagger \hat{a} \,|\, n\rangle = \langle \hat{a}n \,|\, \hat{a}n\rangle = ||\hat{a}n||^2 \geq 0 \\ &= n\langle n \,|\, n\rangle = n||n||^2,\end{aligned}$$

where $||n||^2 \geq 0$ and therefore $n \geq 0$. In addition, from the commutation relations we get

$$\begin{aligned}\hat{a}\hat{N}_a|n\rangle &= n\hat{a}|n\rangle, \\ &= \hat{a}\hat{a}^\dagger \hat{a}|n\rangle = \left[\hat{I} + \hat{a}^\dagger \hat{a}\right]\hat{a}|n\rangle = \hat{a}|n\rangle + \hat{N}_n \hat{a}|n\rangle.\end{aligned}$$

After rearranging these terms, we obtain

$$\hat{N}_a \hat{a}|n\rangle = (n-1)\hat{a}|n\rangle \propto |n-1\rangle.$$

From this we see that

$$|n\rangle \in \text{Ker } (\hat{N}_a - n\hat{I}) \implies \hat{a}|n\rangle \in \text{Ker } (\hat{N}_a - (n-1)\hat{I}),$$

however $n-1 \geq 0$, therefore there must exist $n_0$ such that $\hat{a}|n_0\rangle = 0$. Using the ladder operator relations, we have

$$\hat{a}|n_0\rangle = \sqrt{n_0}|n_0 - 1\rangle.$$

If $n_0 > 0$, then $\sqrt{n_0} \neq 0$, which implies $|n_0 - 1\rangle = 0$. This is inconsistent, as $|n_0 - 1\rangle$ would collapse the entire Fock space to the zero vector, violating the definition of the Fock space. Hence, the only consistent value is $n_0 = 0$.

Overall, the eigenvalues $n$ of the operator $\hat{N}_a$ must be the set of non-negative whole numbers $\{0, 1, 2, \cdots\}$.

$\boxed{2.}$ From the definition of $\hat{O}$, we know that it must act on the vacuum $|0_a\rangle$ to produce the state $|n\rangle$:

$$|n\rangle = \hat{O}|0_a\rangle.$$

Since the state $|n\rangle$ is obtained by applying $(\hat{a}^\dagger)^n$ to the vacuum, we propose that $\hat{O}$ is proportional to $(\hat{a}^\dagger)^n$. Specifically, we compute:

$$(\hat{a}^\dagger)^n|0_a\rangle = \sqrt{n!}\,|n\rangle.$$

Thus, to ensure the correct normalization, we identify

$$\hat{O} = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n.$$

$\boxed{3.}$ Here we use the approach of staight computation with the use of the commutation relations and the properties of $|n\rangle$ states:

$$\begin{aligned}\langle n \,|\, (\hat{a} - \hat{a}^\dagger)^2 \,|\, n\rangle &= \langle n \,|\, (\hat{a})^2 + (\hat{a}^\dagger)^2 - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} \,|\, n\rangle \\ &= \langle n \,|\, (\hat{a})^2 \,|\, n\rangle + \langle n \,|\, (\hat{a}^\dagger)^2 \,|\, n\rangle - 2\langle n \,|\, \hat{a}^\dagger\hat{a} \,|\, n\rangle - \langle n \,|\, [\hat{a}, \hat{a}^\dagger] \,|\, n\rangle \\ &= 0 + 0 - 2\langle n \,|\, \hat{N}_n \,|\, n\rangle - \langle n \,|\, \hat{I} \,|\, n\rangle = -2n - 1\end{aligned}$$

## 0.2 Unitary canonical transformation and coherent states

$\boxed{1.}$ Firstly, we derive the Campbell identity, which states that

$$e^{\hat{X}}\hat{Y}e^{-\hat{X}} = \hat{Y} + [\hat{X}, \hat{Y}] + \frac{1}{2!}[\hat{X}, [\hat{X}, \hat{Y}]] + \frac{1}{3!}[\hat{X}, [\hat{X}, [\hat{X}, \hat{Y}]]] + \cdots.$$

This can be proven by defining the operator function dependent on the parameter $t$:

$$\hat{f}(\hat{Y}; t) \coloneqq e^{t\hat{X}}\hat{Y}e^{-t\hat{X}}.$$

We take the derivative:

$$\frac{\partial}{\partial t}\hat{f}(\hat{Y}; t) = \hat{X}e^{t\hat{X}}\hat{Y}e^{-t\hat{X}} - e^{t\hat{X}}\hat{Y}e^{-t\hat{X}}\hat{X} = [\hat{X}, e^{t\hat{X}}\hat{Y}e^{-t\hat{X}}] = [\hat{X}, \hat{f}(\hat{Y}; t)].$$

We can integrate this equation with respect to $t$ by using the boundary condition $\hat{f}(\hat{Y}; 0) = \hat{Y}$, and we formally obtain

$$\hat{f}(\bullet; t) = e^{t[\hat{X}, \bullet]} \equiv \sum_{n=0}^{\infty} \frac{1}{n!} t^n [\hat{X}, \bullet]^n,$$

where we define

$$[\hat{X}, \bullet]^n \coloneqq \underbrace{[\hat{X}, [\hat{X}, \cdots, [\hat{X}}_{n}, \bullet] \cdots]].$$

We obtain the desired outcome by taking $t = 1$.

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In our case, we deal with operators $\hat{a}$ and $\hat{a}^\dagger$ and their linear combinations. Their commutator is equal to the identity operator, which trivially commutes with all other operators. Therefore, all $[\hat{X}, \bullet]^n$ vanish for $n > 1$, and we get

$$\hat{S}^{-1}\hat{a}\hat{S} = e^{\hat{X}}\hat{a}e^{-\hat{X}} = \hat{a} + [\hat{X}, \hat{a}] = \hat{a} + [c^*\hat{a} - c\hat{a}^\dagger, \hat{a}] = \hat{a} + 0 - c[\hat{a}^\dagger, \hat{a}] = \hat{a} + c = \hat{b}.$$

$\boxed{2.}$ Here we straightforwardly obtain:

$$\hat{b}|0_b\rangle = \hat{S}^{-1}\hat{a}\hat{S}\hat{S}^{-1}|0_a\rangle = \hat{S}^{-1}\hat{a}|0_a\rangle = 0,$$

and therefore $|0_b\rangle$ is the vacuum state of $\hat{b}$. On the other hand, we have:

$$\hat{a}|0_b\rangle = \left(\hat{b} - c\right)|0_b\rangle = 0 - c|0_b\rangle = -c|0_b\rangle.$$

$\boxed{3.}$ Again using the fact that we deal with operators $\hat{a}$ and $\hat{a}^\dagger$ and their linear combinations, and therefore their commutator is equal to the identity operator, which trivially commutes with all other operators, we can use the simplified BCH formula

$$e^{\hat{X}}e^{\hat{Y}} = e^{\hat{X}+\hat{Y}+\frac{1}{2}[\hat{X},\hat{Y}]} \implies e^{\hat{X}+\hat{Y}} = e^{\hat{X}}e^{\hat{Y}}e^{-\frac{1}{2}[\hat{X},\hat{Y}]}.$$

We obtain

$$\begin{aligned}|0_b\rangle = \hat{S}^{-1}|0_a\rangle &= e^{c^*\hat{a}-c\hat{a}^\dagger}|0_a\rangle = e^{-c\hat{a}^\dagger}e^{c^*\hat{a}}e^{\frac{1}{2}[c\hat{a}^\dagger, c^*\hat{a}]}|0_a\rangle \\ &= e^{-c\hat{a}^\dagger}e^{c^*\hat{a}}e^{-\frac{|c|^2}{2}}|0_a\rangle = e^{-\frac{|c|^2}{2}}e^{-c\hat{a}^\dagger}|0_a\rangle,\end{aligned}$$

where we have used the spectral form of operator $e^{c^*\hat{a}}$:

$$e^{c^*\hat{a}}|0_a\rangle = e^{c^*\lambda_{|0_a\rangle}}|0_a\rangle = e^{c^*\cdot 0}|0_a\rangle = \hat{I}|0_a\rangle = |0_a\rangle.$$

Using this result we have

$$\begin{aligned}\langle 0_b \,|\, \hat{N}_a \,|\, 0_b\rangle &= e^{-|c|^2}\langle e^{-c\hat{a}^\dagger}0_a \,|\, \hat{a}^\dagger\hat{a} \,|\, e^{-c\hat{a}^\dagger}0_a\rangle = e^{-|c|^2}\langle 0_a \,|\, e^{-c^*\hat{a}}\hat{a}^\dagger\hat{a}e^{-c\hat{a}^\dagger} \,|\, 0_a\rangle \\ &= e^{-|c|^2}\langle 0_a \,|\, e^{-c^*\hat{a}}\hat{a}^\dagger e^{-c\hat{a}^\dagger}e^{+c\hat{a}^\dagger}\hat{a}e^{-c\hat{a}^\dagger} \,|\, 0_a\rangle \\ &= e^{-|c|^2}\langle 0_a \,|\, e^{-c^*\hat{a}}\hat{a}^\dagger e^{-c\hat{a}^\dagger}[\hat{a} - c\hat{I}] \,|\, 0_a\rangle = -ce^{-|c|^2}\langle 0_a \,|\, e^{-c^*\hat{a}}\hat{a}^\dagger e^{-c\hat{a}^\dagger} \,|\, 0_a\rangle \\ &= -ce^{-|c|^2}\langle 0_a \,|\, e^{-c^*\hat{a}}\hat{a}^\dagger e^{+c^*\hat{a}}e^{-c^*\hat{a}}e^{-c\hat{a}^\dagger} \,|\, 0_a\rangle \\ &= -ce^{-|c|^2}\langle 0_a \,|\, [\hat{a}^\dagger - c^*\hat{I}]e^{-c^*\hat{a}}e^{-c\hat{a}^\dagger} \,|\, 0_a\rangle = |c|^2e^{-|c|^2}\langle 0_a \,|\, e^{-c^*\hat{a}}e^{-c\hat{a}^\dagger} \,|\, 0_a\rangle \\ &= |c|^2e^{-|c|^2}\langle e^{-c\hat{a}^\dagger}0_a \,|\, e^{-c\hat{a}^\dagger}0_a\rangle = |c|^2\langle 0_b \,|\, 0_b\rangle = |c|^2,\end{aligned}$$

where we have used the Campbell identity several times.

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